I-195/SR-112 Bridge – Structural Monitoring Plan – Proposal

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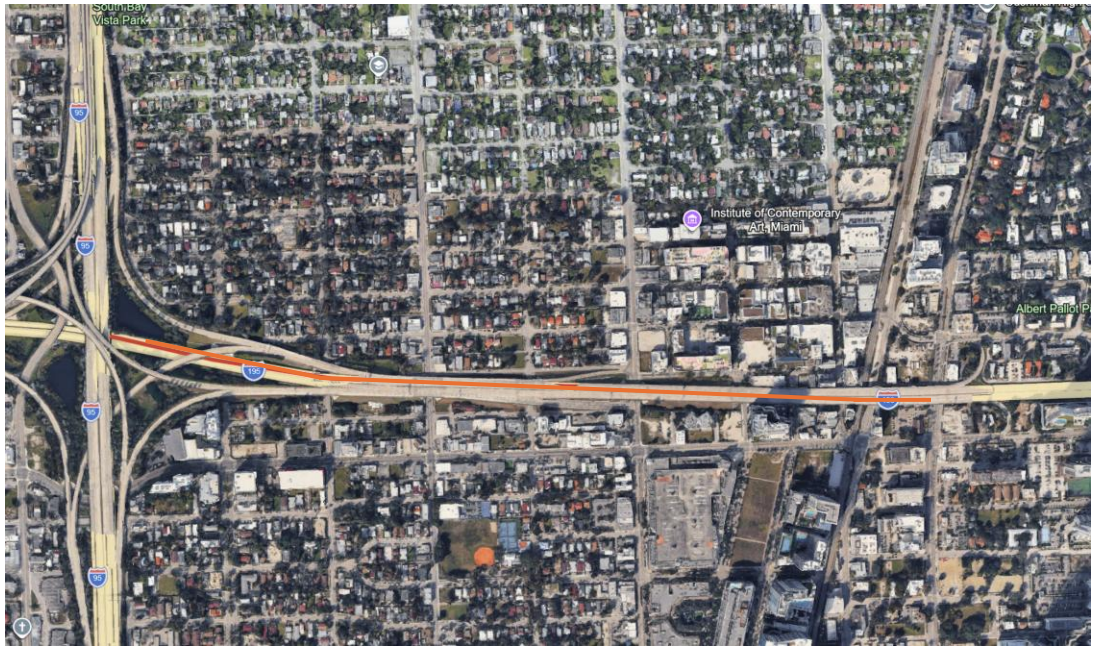
I-195/SR-112 Bridge, Miami-Dade County

The I-195 Bridge between I-95 and US-1 is one of the most critical bridges in Miami, with an ADT of 151,500 vehicles per day. It is currently classified as being in poor condition based on the NBI report.

Developing a structural health monitoring plan is not a luxury, but an urgent necessity to prevent potential loss of life and significant economic consequences.



Environmental Factors

Environmental / operational factor	Why it matters
High daily traffic	Repeated live-load cycles increase fatigue demand.
Urban setting	Inspection access is more difficult because the bridge crosses roadway and railway infrastructure.
Railroad and highway below	Falling debris, deterioration, or emergency closures could affect multiple transportation systems.
Miami-Dade coastal climate	Heat, humidity, and chloride exposure may accelerate corrosion and material degradation.
Skewed / complex geometry	The bridge has a high skew value in the NBI data, which may create nonuniform load distribution and localized stress concentrations.

Item	Bridge information
Structure number	870313
State	Florida
County	Miami-Dade County
Facility carried	I-195 / SR-112
Feature crossed	SR-5 and FEC Railroad
Location	Between I-95 and US-1
Year built	1961
Year reconstructed	2000
Average daily traffic	151,500 vehicles/day
Truck traffic	5%
Traffic lanes on bridge	6
Traffic lanes under bridge	15
Structure length	432.0 m
Maximum span length	33.9 m
Deck width	32.5 m
Roadway width	29.9 m
Bridge condition	Poor
Lowest rating	3

Materials

Despite the NBI report of using steel girder, the structural system of the bridge is concrete girder

NBI item	Value	Interpretation
Structure Kind 043A	3	Steel
Structure Type 043B	2	Stringer / multi-beam or girder system
Deck Structure Type 107	1	Concrete cast-in-place deck
Main unit spans	1	
Approach spans	19	

Condition

The most critical element are the concrete girders, which form the superstructure of the bridge

Potential Deterioration Mechanisms / Risks	
Possible fatigue cracking	High traffic volume creates repeated stress cycles.
Localized connection deterioration	Girder-to-diaphragm and bearing zones may be fatigue-sensitive.
Impact or clearance-related vulnerability	The bridge crosses road and rail infrastructure, so underside elements may be exposed to impact risk.
Potential nonuniform load distribution	Skew and complex geometry may create localized overstress.

Component	NBI Rating	General interpretation	Relevance to monitoring
Deck	6	Satisfactory	Some minor deterioration may exist, but it is not the main concern.
Superstructure	3	Serious condition	Primary concern; monitoring should focus here.
Substructure	7	Good	Not the main concern but should still be visually checked.
Channel	N	Not applicable	Bridge does not require a waterway channel rating.
Overall bridge condition	Poor	Controlled by lowest major component rating	The poor classification is driven by the superstructure.
Lowest rating	3	Serious	Strong justification for Phase 1 monitoring.

Economic and Life-Safety Justifications

A peer-reviewed U.S. bridge failure-rate study estimated 87–222 bridge collapses/failures per year, with an expected value of 128 per year. The 2025 FHWA NBI lists 624,193 U.S. bridges. So:

$$P \approx \frac{128}{624{,}193} = 0.000205$$

That is about:
0.0205% per bridge-year, or about 1 in 4,877 per year.

Scenario	Annual probability	65-year probability
Low	0.01%	0.90%
Expected	0.02%	1.32%
High	0.04%	2.29%

Using Sensitivity analysis the P(x=65 years) = 1.485%. (Prob. Of failure)

Probability of Failure at 65 years	Life-Loss Cost/Fatality ($M)	Estimated Fatalities	Total Life-Loss Cost ($M)	Economic Cost ($M/Year)	First Year Cost ($B)	Second Year Cost ($B)
1.49%	14.2	14.5	205.9	1340	1.5	2.8

Structural Health Monitoring System

Assumptions

- Phase 1 focuses on detection, not full damage diagnosis.
- The main damage scenarios are fatigue cracking and corrosion-related section loss.
- Normal traffic will be used as the primary excitation source.
- Temperature effects must be measured to avoid false damage alarms.
- Sensor placement will prioritize accessible, fatigue-sensitive, and repair-sensitive superstructure locations.

Damage definition: Monitoring will focus on the girder, since NBI inspection data identifies it as the most critical structural component. Following Farrar's approach, damage must be defined early so the SHM system is aligned with its purpose. In this Phase I plan, damage is defined as the initiation and progression of fatigue cracking in the concrete girder. The SHM system is therefore designed to detect anomalies or changes in girder response that could indicate developing damage. In later phases, the system can be refined to track specific crack locations, propagation, and other local damage indicators.

Decision	Proposed choice	Justification
Monitoring priority	Superstructure	NBI superstructure rating is the controlling concern.
Main damage scenario	Fatigue cracking and corrosion-related section loss in concrete girders	Common concerns for this type of bridges
Monitoring type	Permanent Phase 1 sensors + targeted inspections	Practical within one year and suitable for asset managers.
Data strategy	Baseline comparison	Compare future measurements against normal traffic/temperature behavior.
Sensor quantity	Limited number of sensors	Bridges are large, unique systems, so monitoring must be practical with a cost-effective number of sensors.
Excitation source	Normal traffic	The bridge should be monitored during normal operation, without interrupting traffic.

Monitoring System

The system will use a limited number of accelerometers, strain gauges, and temperature sensors to establish baseline vibration and live-load strain behavior under normal traffic. This approach follows the SHM principles discussed by Farrar, where bridge monitoring must define a limited number of priority damage scenarios and account for environmental and operational variability. The primary damage scenarios for this bridge are fatigue cracking and corrosion-related section loss. Accelerometers will track changes in vibration response, strain gauges will measure fatigue-sensitive live-load strain ranges, and temperature sensors will help separate true structural changes from thermal effects. Targeted NDT, including ultrasonic thickness testing and magnetic particle testing, will be used to confirm whether abnormal sensor trends correspond to physical deterioration.

Sensor System

Sensor type	Purpose	Justification
Accelerometers	Measure vibration response, frequency content, and dynamic changes	Useful for detecting changes in global bridge behavior.
Strain gauges	Measure live-load strain and strain ranges in steel girders	Directly relevant to fatigue cracking and stress demand under traffic.
Temperature sensors	Measure thermal effects on strain and vibration response	Needed because temperature changes can look like structural damage if not corrected.

Recommended Features

Feature	Sensor	Damage relevance
Natural frequency trend	Accelerometers	Global stiffness changes
Acceleration RMS / peak response	Accelerometers	Abnormal dynamic response
Strain range under traffic	Strain gauges	Fatigue demand
Daily and seasonal temperature trends	Temperature sensors	Environmental correction
Temperature-normalized frequency	Accelerometer + temperature	Reduces false alarms
Strain anomalies at details	Strain gauges	Local fatigue-sensitive behavior

Monitoring System - Plan

Phase I - Sensor Plan		
Sensor	**Quantity**	**Location**
Accelerometers	4–6	Midspan, quarter points, and one support region on selected spans
Strain gauges	6–8	Critical girders, connections, repair-sensitive areas
Temperature sensors	3–4	Steel girder surface, deck underside, shaded/exposed locations
Data acquisition unit	1–2	Mounted in protected cabinet near access point

One year Schedule	
Period	**Activity**
Months 1–2	Confirm access, review inspection history, select sensor locations
Months 3–4	Install sensors and collect initial baseline
Months 5–10	Monitor traffic-induced vibration, strain range, and temperature effects
Months 11–12	Analyze trends, perform targeted NDT, issue Phase 1 recommendation

Sensor Layout			
Location	**Sensor type**	**Measurement**	**Justification**
Midspan of selected critical span	Accelerometer + strain gauge	Maximum live-load vibration and bending strain	Midspan is typically sensitive to flexural response.
Near Span 20 / repair-sensitive region, if access confirms this area	Strain gauge + accelerometer	Local live-load response and abnormal vibration	Useful if past impact/repair concerns exist.
Near girder ends / support regions	Strain gauge or displacement observation point	Support restraint, bearing-related response, local stress changes	Bearing restraint can change measured response.
Near diaphragms, connections, or welded attachments	Strain gauge	Local fatigue-sensitive strain range	Fatigue cracks often initiate near details with stress concentration.
One exposed location and one shaded location	Temperature sensors	Temperature gradients and thermal correction	Prevents false damage alarms caused by daily

Sensor Specifications		
Sensor type	**Baumer model**	**Key specs**
Strain	DST20-A102P	±1000 μm/m, 1.0 mV/V, IP65, stainless steel, <0.5% non-linearity, >10 million cycles
Temperature	TE2	Compact RTD temperature sensor, rugged design, suitable for thermal compensation
Acceleration	GIM700DR	Integrated triaxial accelerometer, ±6 g, 100 Hz, CANopen/SAE J1939

NTD Method

Ultrasonic thickness testing: used to detect corrosion and local section loss in steel members. It should be applied to girder and connection zones where moisture and deterioration are more likely to accumulate.

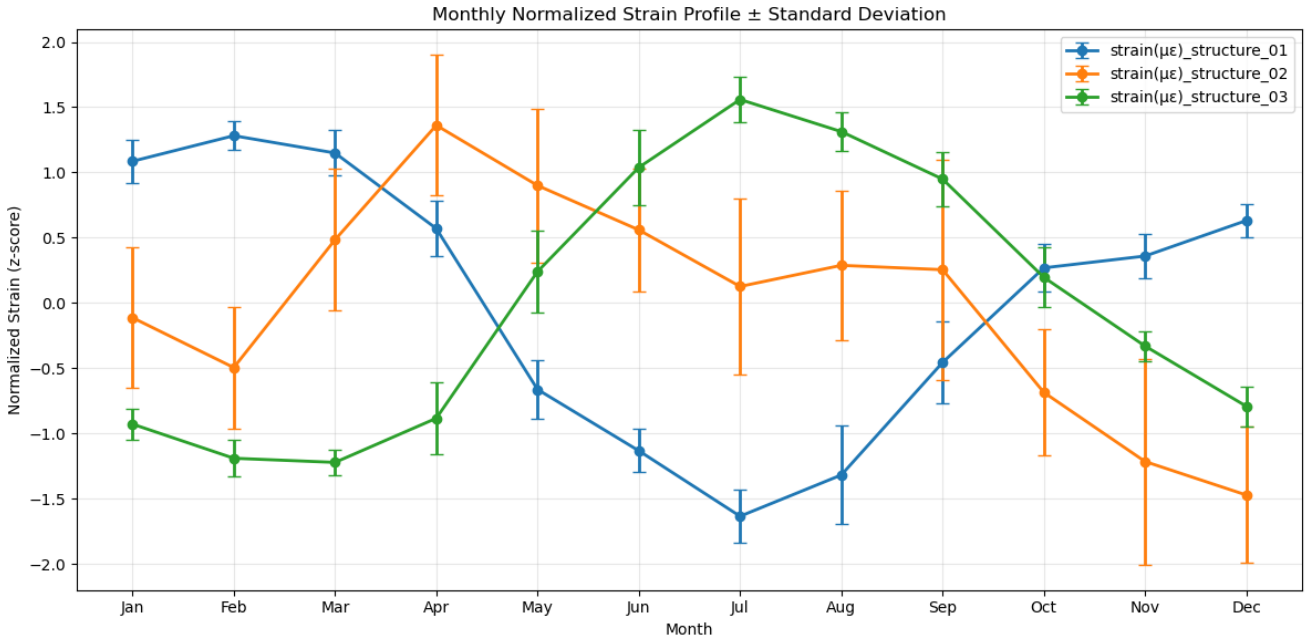
Magnetic particle testing: used to identify surface-breaking fatigue cracks in steel details. It should be focused on welds, stiffeners, diaphragms, and other repair-sensitive or stress-concentration areas.

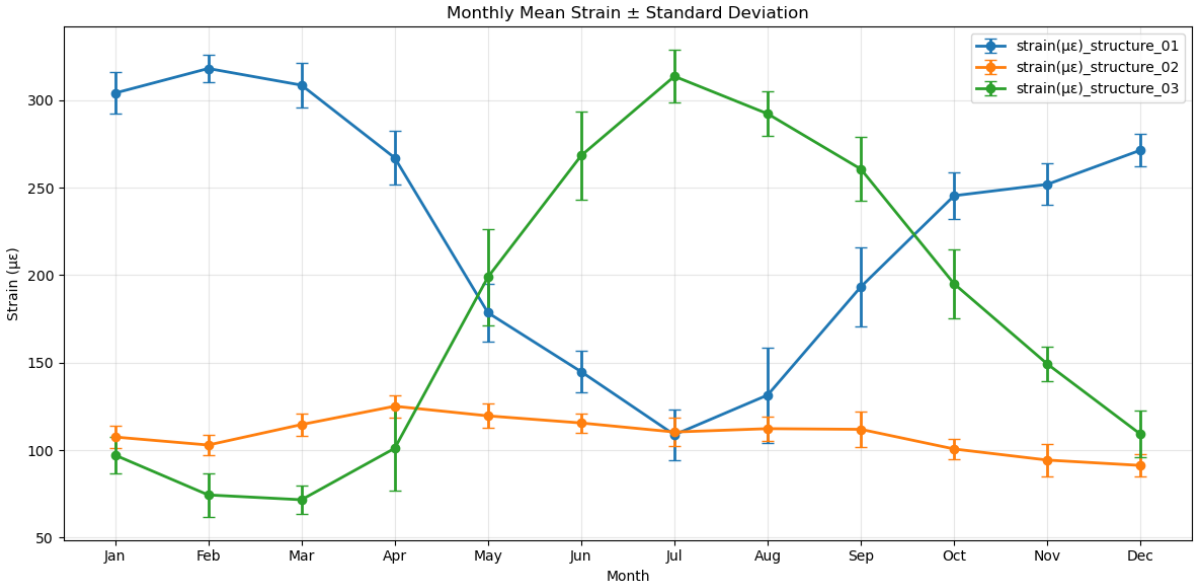
Ground Penetrating Radar is one of the most effective NDT methods because it can quickly screen for deck deterioration, moisture intrusion, voids, and reinforcement layout. If corrosion is a concern, it can be complemented with half-cell potential testing.

Conclusion

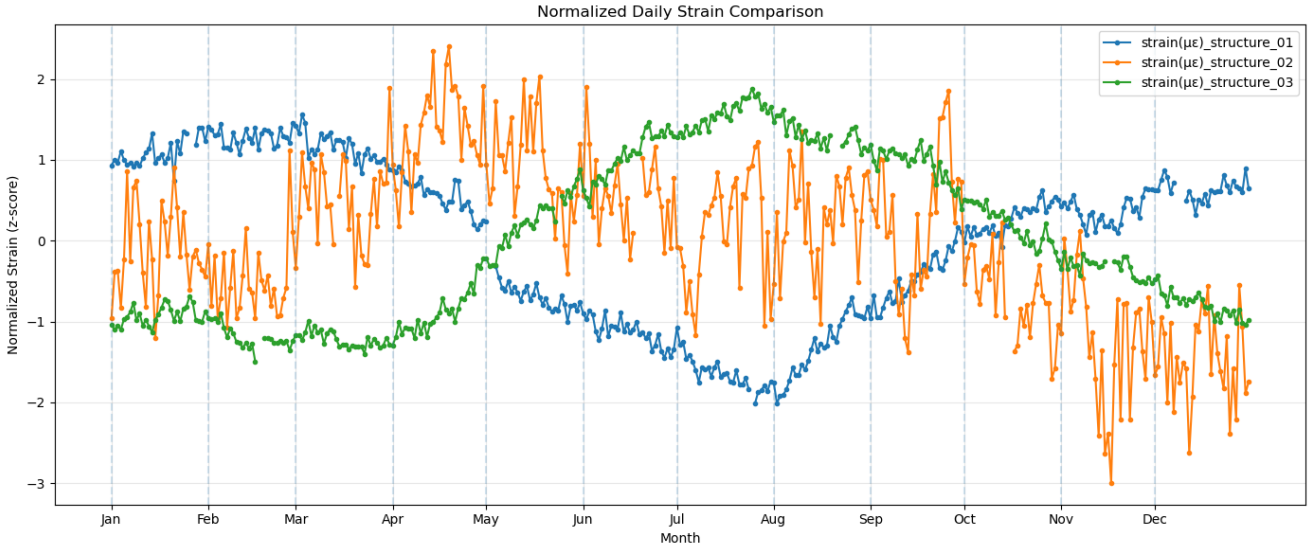
- Bridge 870313 was selected because it is a high-traffic urban bridge with a poor superstructure rating, making it a strong candidate for a Phase 1 monitoring plan.
- The main concern is the concrete superstructure, where likely deterioration mechanisms include cracking, delamination, moisture intrusion, and possible reinforcement corrosion.
- A minimum viable monitoring system using strain gauges, accelerometers, and temperature sensors can be deployed within one year to establish baseline structural behavior and detect abnormal changes over time.
- Targeted NDT, such as Ground Penetrating Radar (GPR) and Ultrasonic Pulse Velocity (UPV), can complement the sensors by helping confirm internal defects, deck deterioration, or other concrete-related damage.
- Overall, this Phase 1 plan is practical, feasible, and useful for asset management, since it supports better inspection prioritization and creates a foundation for future monitoring improvements.

Task 2





Structure	mean	std	min	max	range	cv_%
Strain (Eμ)_structure_01	226.1	71.7	81.8	337.8	256	31.7
Strain (Eμ)_structure_02	108.8	11.9	73.1	137.5	64.4	11.0
Strain (Eμ)_structure_03	178.0	87.0	48.2	341.5	293.3	48.9



Correlation Matrix	strain(Eμ)_structure_01	strain(Eμ)_structure_02	strain(Eμ)_structure_03
strain(Eμ)_structure_01	1	-0.22	-0.95
strain(Eμ)_structure_02	-0.22	1	0.16
strain(Eμ)_structure_03	-0.95	0.16	1

Conclusions

Structures 01 and 03 are strongly affected by seasonal changes, which suggests sensitivity to environmental effects. However, based on strain data alone, it is not possible to confirm the exact structure type.

Structure 02 shows a different behavior, with lower overall variability and a flatter annual pattern, even though its daily readings contain noticeable short-term fluctuations.

Structures 01 and 03 are not just highly correlated; they are highly negatively correlated (r = -0.95). This indicates opposite response trends over time. This may reflect different sensor locations or orientations, but it is not enough evidence to conclude that they belong to the same structure.